

Longleaf Execution Guide — multiomicsGEP

Run the Supervised Bayesian Matrix Factorization project on UNC's Longleaf HPC cluster.

Requirements: R 4.4.0, two packages (ebnm, survival).

What's in This Directory

```
longleaf_setup/
├─ README.md           # This file – the guide you're reading
├─ install_packages.R   # One-time R package installer (ebnm + survival)
├─ run_tests.sl         # SLURM job: run the 105-test suite
├─ run_v2_simulation.sl  # SLURM job: run the V2 monolithic simulation
├─ run_modular_simulation.sl # SLURM job: run the modular simulation
├─ run_demos.sl         # SLURM job: run all 4 demo scripts
└─ logs/               # SLURM writes job output here (stdout + stderr)
```

What are .sl files?

.sl files are **SLURM job scripts** — shell scripts with special #SBATCH comment lines at the top that tell Longleaf how much memory, time, and CPUs your job needs. When you run `sbatch some_script.sl`, Longleaf reads those directives, puts your job in a queue, and runs it on a compute node when resources are available. You don't need to wait at your terminal — the output goes to a log file you can check later.

What are the two simulations?

The project has two ways to run the same algorithm:

- **V2 simulation** (`run_v2_simulation.sl`): Runs `code/Supervised_Bayesian_MF_V2.R`, the single monolithic script that contains everything — the CAVI loop, all update equations, helpers, and visualization. This is the original implementation.
- **Modular simulation** (`run_modular_simulation.sl`): Runs `results/run_modular_simulation.R`, which builds the CAVI loop from the four standalone update modules (`code/update_L.R`, `update_F.R`, `update_beta.R`, `update_tau.R`). Same algorithm, same math, but using the modular code that has its own test suite.

Both produce the same outputs (7 CSV tables + 8 figure pairs in PDF and PNG). You can run either or both — they use different random seeds so results will differ slightly, but the algorithm behavior should be consistent.

What do the other jobs do?

- **Tests** (`run_tests.sl`): Runs the 105-test suite that validates the four modular update scripts. Good to run first to confirm everything works on Longleaf.
 - **Demos** (`run_demos.sl`): Runs the four educational demo scripts that illustrate the mathematical properties of each update (signal recovery, shrinkage behavior, error-in-variables effects, etc.). Output is narrative text — no figures are saved.
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How Longleaf Works (Quick Primer)

Longleaf is a shared compute cluster. You **SSH in** to a login node, but you **don't run heavy computation there** — instead you either:

1. **srun**: Request an interactive session on a compute node (like getting a temporary terminal on a powerful machine). Good for installing packages and quick test runs.
2. **sbatch**: Submit a job script to the queue. Longleaf runs it when resources are free and saves the output to a log file. Good for anything that takes more than a few minutes.

Your files live on a shared filesystem, so both the login node and compute nodes can see the same `/work/users/...` directory.

Phase 1 — First-Time Setup

Do this once when you first set up the project on Longleaf.

```
# 1. SSH into Longleaf from your Mac terminal
ssh <onyen>@longleaf.unc.edu

# 2. Go to your work directory
#   Replace the letters and onyen with yours.
#   Example: if your onyen is "awalther", the path would be
#   /work/users/a/w/awalther
cd /work/users/<first_letter>/<second_letter>/<onyen>

# 3. Clone the repo (one-time)
git clone https://github.com/andrew-walther/multiomicsGEP.git
cd multiomicsGEP

# 4. Create the directories that jobs will write into
#   (git doesn't track empty directories, so we create them manually)
```

```
mkdir -p longleaf_setup/logs \
      results/tables/full_sim results/tables/modular_sim \
      results/figures/full_sim results/figures/modular_sim

# 5. Install R packages in an interactive session
#   srun gives you a shell on a compute node (the login node is not
#   meant for running R). The flags request 4 GB RAM for 30 minutes.
srun -p interact -n 1 --mem=4G -t 00:30:00 --pty bash

# Now you're on a compute node. Load R and install packages:
module add r/4.4.0
Rscript longleaf_setup/install_packages.R

# When it finishes, leave the compute node:
exit
```

After this, `ebnm` and `survival` are installed in your personal R library on Longleaf. You won't need to do this again unless you delete your library or the R module version changes.

Phase 2 — Test Run (Verify Before Full Submission)

Before submitting batch jobs, confirm that R and the code work interactively.

```
# 6. Start an interactive session and run the test suite
srun -p interact --mem=2G --time=00:10:00 --pty bash
module add r/4.4.0
export REPO_ROOT=$(pwd)
Rscript tests/run_tests.R

# You should see:
#   --- tests/test_update_beta.R: 24/24 tests passed ---
#   --- tests/test_update_L.R:    28/28 tests passed ---
#   --- tests/test_update_F.R:    26/26 tests passed ---
#   --- tests/test_update_tau.R:  27/27 tests passed ---
#   FINAL: ... All tests PASSED.

exit
```

If all 105 tests pass, you're good to go. If something fails, check that you ran steps 4 and 5 from Phase 1.

Phase 3 — Submit Jobs

Now you can submit batch jobs. These run in the background — you can log out and come back later.

```
# 7. Submit whichever jobs you want (from the repo root directory)
sbatch longleaf_setup/run_tests.sl           # ~30 seconds
sbatch longleaf_setup/run_v2_simulation.sl    # ~5 minutes
sbatch longleaf_setup/run_modular_simulation.sl # ~5 minutes
sbatch longleaf_setup/run_demos.sl           # ~2 minutes

# Each sbatch command prints a job ID like:
#   Submitted batch job 12345678

# 8. Check on your jobs
squeue -u <onyn>          # shows your running/queued jobs
squeue -u <onyn> | wc -l  # quick count (subtract 1 for the header line)
```

Tip: You don't need to submit all four. If you just want the modular simulation, run only `sbatch longleaf_setup/run_modular_simulation.sl`.

Phase 4 — Check Results

When `squeue` shows no more jobs (or you get an email if configured), check the output.

```
# 9. View job logs (stdout from your R scripts)
cat longleaf_setup/logs/mgep_tests_*.out      # test results
cat longleaf_setup/logs/mgep_v2sim_*.out      # V2 simulation console output
cat longleaf_setup/logs/mgep_modsim_*.out     # modular simulation console output
cat longleaf_setup/logs/mgep_demos_*.out     # demo narrative output

# Check for errors (empty = good)
cat longleaf_setup/logs/mgep_v2sim_*.err

# Verify output files were created
ls results/tables/full_sim/                  # should see 7 CSVs
ls results/figures/full_sim/                 # should see 8 PDF + 8 PNG files
ls results/tables/modular_sim/               # should see 7 CSVs
ls results/figures/modular_sim/              # should see 8 PDF + 8 PNG files
```

Expected outputs per simulation

Each simulation produces:

Directory	Contents
results/tables/*/factor_summary_table.csv	Per-factor beta, log-rank p, sparsity, PVE
results/tables/*/beta_comparison_table.csv	Estimated vs. true betas with posterior SD
results/tables/*/cindex_comparison.csv	C-index: supervised loadings vs. PCA
results/tables/*/convergence_history.csv	RMSE and ELBO per iteration
results/tables/*/top_features_GEP[1-5].csv	Top 10 features per factor
results/tables/*/loading_correlation_matrix.csv	True vs. estimated loading correlations
results/tables/*/ph_test_results.csv	Proportional hazards test
results/figures/*/fig[1-8]_*.pdf + .png	RMSE trace, ELBO, beta comparison, heatmap, KM curves, signal recovery, correlation heatmap, tau distribution

Phase 5 — Get Results to Your Mac

Run this **on your Mac** (not on Longleaf) to copy the results back:

```
# 10. From your Mac terminal:
scp -r <onyen>@longleaf.unc.edu:/work/users/<first>/<second>/<onyen>/multiomicsGEP/results/ \
~/GithubProjects/multiomicsGEP/results/
```

This copies the entire results/ directory (tables, figures, logs) to your local repo, where you can open the PDFs and CSVs directly.

Quick Reference

SLURM scripts at a glance

Script	Job Name	Mem	Time	What it runs
run_tests.sl	mgep_tests	2g	10 min	Test suite (105 tests)
run_v2_simulation.sl	mgep_v2sim	4g	1 hr	V2 monolithic simulation
run_modular_simulation.sl	mgep_modsim	4g	1 hr	Modular simulation
run_demos.sl	mgep_demos	2g	30 min	All 4 demo scripts

All scripts log to longleaf_setup/logs/ and must be submitted from the repo root.

Common commands

What	Command
Submit a job	<code>sbatch longleaf_setup/run_tests.sl</code>
Check your queue	<code>squeue -u <onyen></code>
Cancel a job	<code>scancel <job_id></code>
Cancel all your jobs	<code>scancel -u <onyen></code>
View job output	<code>cat longleaf_setup/logs/mgep_tests_*.out</code>
View job errors	<code>cat longleaf_setup/logs/mgep_tests_*.err</code>
Interactive session	<code>srun -p interact --mem=2G -t 00:10:00 --pty bash</code>
Check available R	<code>module avail r</code>
Load R	<code>module add r/4.4.0</code>

Troubleshooting

- **“Cannot find repo root”** error in R: You submitted the job from the wrong directory. `cd` to `multiomicsGEP/` before running `sbatch`.
- **Package not found:** Re-run Phase 1, step 5 (install packages).
- **Job stuck in queue:** Longleaf is busy. Check `squeue -u <onyen>` — status `PD` means pending (waiting for resources), `R` means running.
- **Out of memory:** Increase `--mem` in the `.sl` file (e.g., `--mem=8g`).
- **Out of time:** Increase `-t` (e.g., `-t 02:00:00` for 2 hours).